DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001-2015 Certified Institution)

Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru - 560 111. India

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<tr><td>DEPARTMENT</td><td colspan="5">ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING</td></tr>
<tr><td colspan="6">Semester: VII   Course Title: Data Security & Privacy Lab   Course Code: 22AI73</td></tr>
<tr><td>PROGRAM QUESTION</td><td colspan="5">Implement digital signature generation and verification using RSA,Use cryptography or PyCrypto libraries. b) Simulate authentication and authorization using a web application</td></tr>
<tr><td>STUDENT NAME</td><td colspan="2">Udayaraj</td><td>USN</td><td colspan="2">1DS22AI057</td></tr>
<tr><td>LAB MARKS</td><td>Program Execution (10)</td><td>Source Code (10)</td><td>Result (5)</td><td>Innovation/ Extra Features (5)</td><td>Total</td></tr>
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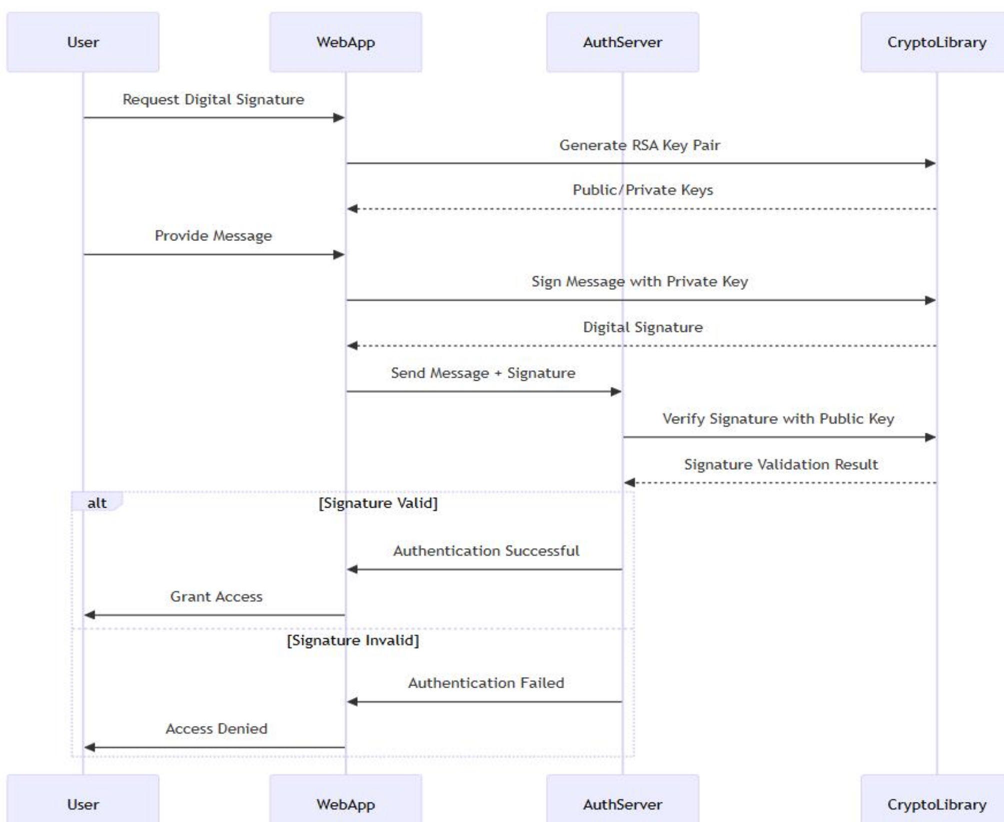
<tr><td>UML / SEQUENCE DIGRAM</td><td colspan="5">
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<tr><td>COURSE COORDINATOR</td><td colspan="5">Dr ARUNA M G & Prof ENSTEIH SILVIA</td></tr>
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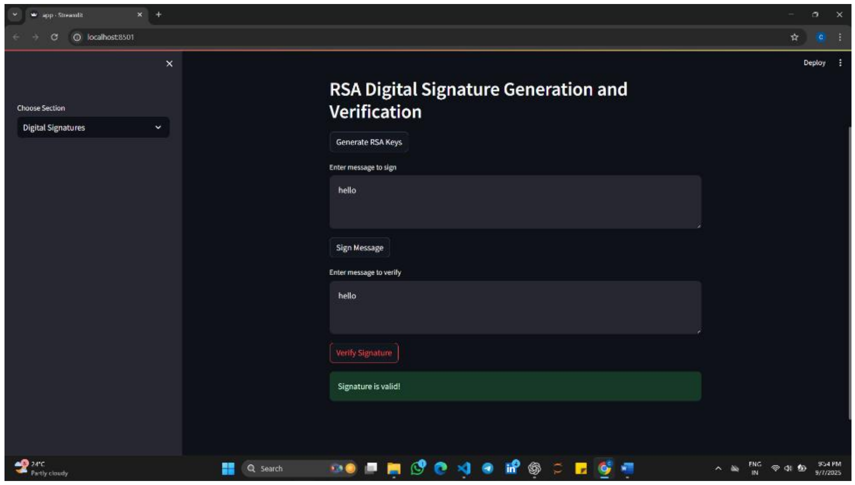
Code for Program 8:

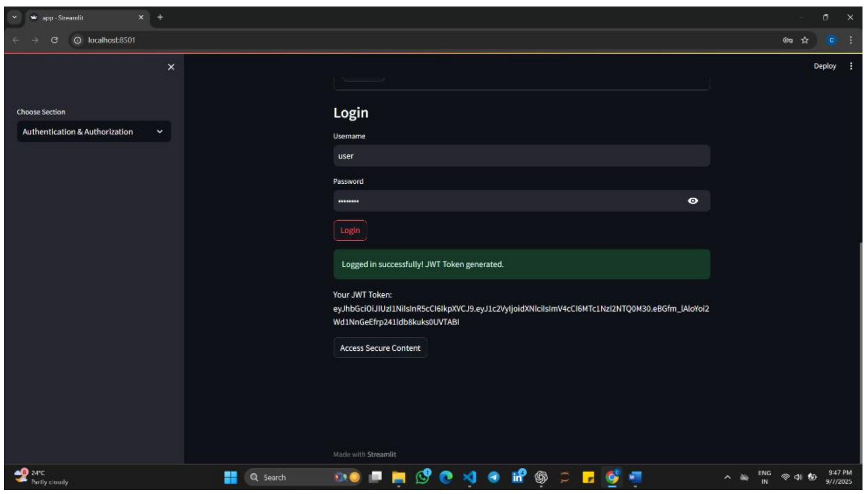
8) Implement digital signature generation and verification using RSA. Use cryptography or PyCrypto libraries. b) Simulate authentication and authorization using a web application

```
import streamlit as st
import jwt
from cryptography.hazmat.primitives import hashes
from cryptography.hazmat.primitives.asymmetric import rsa, padding
import datetime

SECRET_KEY = "your-secret-key"

if 'state' not in st.session_state:
    st.session_state.update({'private_key': None, 'public_key': None, 'signature': None, 'token': None, 'users': []})

st.title("Digital Signatures & Auth Demo")
option = st.sidebar.selectbox("Choose", ["Digital Signatures", "Auth & Authorization"])

if option == "Digital Signatures":
    if st.button("Generate RSA Keys"):
        key = rsa.generate_private_key(public_exponent=65537, key_size=2048)
        st.session_state.update({'private_key': key, 'public_key': key.public_key()})
        st.success("Keys generated!")

    if st.session_state.private_key:
        msg = st.text_area("Message to sign")
        if st.button("Sign"):
            st.session_state.signature = st.session_state.private_key.sign(
                msg.encode(), padding.PSS(mgf=padding.MGF1(hashes.SHA256()), salt_length=padding.PSS.MAX_LENGTH),
hashes.SHA256())
            st.success("Signed!")

        verify = st.text_area("Verify message")
        if st.button("Verify") and st.session_state.signature:
            try:
                st.session_state.public_key.verify(
                    st.session_state.signature, verify.encode(),
                    padding.PSS(mgf=padding.MGF1(hashes.SHA256()), salt_length=padding.PSS.MAX_LENGTH),
                    hashes.SHA256())
                st.success("Valid signature!")
            except:
                st.error("Invalid signature!")

else:
    with st.expander("Register User"):
        u = st.text_input("Username")
        p = st.text_input("Password", type="password")
        c = st.text_input("Confirm", type="password")
        if st.button("Register"):
            if p != c:
                st.error("Mismatch")
            elif any(x['username'] == u for x in st.session_state.users):
                st.error("Exists")
            else:
                st.session_state.users.append({'username': u, 'password': p})
                st.success("Registered!")
```

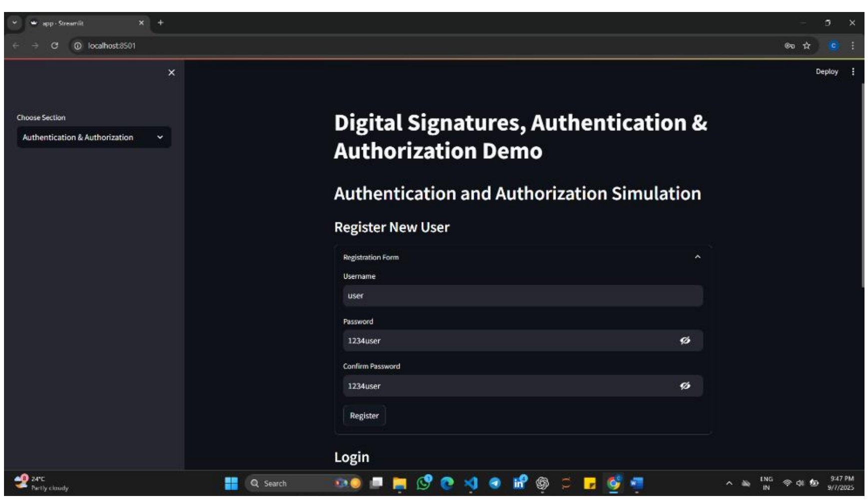
RESULT:



app · Streamlit
localhost:8501
Deploy
Choose Section
Digital Signatures
RSA Digital Signature Generation and Verification
Generate RSA Keys
Enter message to sign
hello
Sign Message
Enter message to verify
hello
Verify Signature
Signature is valid!



app · Streamlit
localhost:8501
Deploy
Choose Section
Authentication & Authorization
Login
Username
user
Password
Login
Logged in successfully! JWT Token generated.
Your JWT Token:
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VyIjoidXNlciIsImV4cCI6MTc1NzI2NTQ0M30.eBGfm_lAloYoi2
Wd1NnGeEfrp241ldb8kuks0UVTABI
Access Secure Content
Made with Streamlit



app · Streamlit
localhost:8501
Deploy
Choose Section
Authentication & Authorization
Digital Signatures, Authentication & Authorization Demo
Authentication and Authorization Simulation
Register New User
Registration Form
Username
user
Password
1234user
Confirm Password
1234user
Register
Login